Real-time Data Processing and Intelligent Monitoring System of Ferrite Magnetic Sensors under Edge Computing Architecture

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***Abstract*—In the field of modern industrial monitoring, the traditional centralized computing architecture is difficult to meet the real-time and reliability requirements of ferrite magnetic sensors in complex industrial environments. This study proposes a real-time data processing and intelligent monitoring system for ferrite magnetic sensors based on edge computing architecture. The system adopts a four-level modular design. The edge perception layer deploys lightweight embedded nodes (ARM Cortex-M7 + ferrite sensor), the edge computing layer builds a distributed processing unit based on the KubeEdge edge containerization framework, the cloud collaboration layer dynamically updates the edge model parameters through federated learning, and the visualization platform integrates the time series database (TDengine) and the low-code dashboard (Node-RED). The maximum delay of the edge computing architecture is 395 milliseconds, the minimum delay is 217 milliseconds, and the average data processing throughput is about 10549 items/second. The system improves the monitoring performance of ferrite magnetic sensors.**

Keywords—edge computing, intelligent monitoring, data processing throughput, f1 value.

I. Introduction

Ferrite magnetic sensors monitor the changes of magnetic fields in the environment in real time, providing key data support for industrial monitoring. However, the traditional centralized computing architecture is difficult to meet the requirements of industrial monitoring for high real-time and high reliability when processing real-time data of ferrite magnetic sensors. This paper proposes a real-time data processing and intelligent monitoring system for ferrite magnetic sensors based on edge computing architecture. Through a four-level modular design, the system realizes full-link optimization from data acquisition, processing, analysis to visualization. Through comparative experiments, the advantages of edge computing architecture in improving the real-time and accuracy of the system are verified. Finally, the system proposed in this paper is technically verified and robustness analyzed. This paper first introduces the research background and related work; explains the system architecture design, the implementation of functions at each level, and the design of the visualization platform; then evaluates the system performance through the experimental part to verify the effectiveness of the system; finally, summarizes the whole article and proposes future research directions.

II. Related Work

As an emerging computing paradigm, edge computing has attracted widespread attention. Ding et al. [1] proposed a horizontal multi-host architecture to optimize the computing latency of the mobile edge computing (MEC) system. In order to optimize the transmission latency of the MEC system and solve the problem of stragglers in multi-host parallel computing, they proposed a multi-connection master-slave multi-host architecture. For the performance evaluation of the MEC system, a multi-host MEC simulation platform based on an open source library was built. The proposed horizontal multi-host MEC architecture can effectively improve the computing latency performance; the proposed multi-connection master-slave multi-host MEC architecture can effectively alleviate the stragglers problem and improve the transmission latency performance. Tu et al. [2] considered the introduction of edge computing technology and proposed a satellite IoT edge computing architecture based on semantic communication. They also analyzed the characteristics and typical application scenarios of the architecture and introduced the challenges that the satellite IoT based on semantic communication will face in the future. Yang et al. [3] explored the optimization strategy of 5G edge computing architecture based on user plane function (UPF). By analyzing the role of UPF in 5G networks and the advantages of edge computing, a new architecture design and optimization method was proposed to improve network performance and user experience. Liu et al. [4] used the edge computing capability of the distribution transformer terminal to perceive the equivalent generalized load (source) of each substation, including distributed power sources (DG) and electric vehicles, based on the physical quantity information of the monitored distribution transformer. On this basis, edge computing was used to perform on-site control of the grid-connected operating conditions of electric vehicles connected to the distribution network in each substation. In view of the current status and problems of smart tunnel monitoring, Hu [5] designed a smart tunnel monitoring system based on edge computing architecture, focusing on the system architecture design and construction process. The system can effectively eliminate the safety hazards of traditional tunnels and improve the operational efficiency of tunnels. Siriwardhana et al. [6] conducted a comprehensive analysis of the architecture, applications, and technologies of mobile augmented reality combined with 5G mobile edge computing (MEC), and discussed the current research progress and future development direction. Al-Doghman et al. [7] reviewed the application scenarios, technical architecture, and challenges of AI-driven secure microservices in edge computing, and proposed corresponding solutions. Zangana et al. [8] conducted a comprehensive analysis of decentralized and collaborative computing models in distributed edge computing. Naouri et al. [9] proposed a novel framework to improve the performance and resource utilization of MEC systems by optimizing task offloading strategies. The framework combines deep learning and optimization

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algorithms to dynamically make task offloading decisions. Abdellatif et al. [10] proposed a framework that combines edge computing and blockchain technology, which performs data preprocessing on edge devices and secure data exchange using blockchain technology. Carvalho et al. [11] analyzed the key technologies, application scenarios, and architecture models of edge computing. Fazeldehkordi and Grønli [12] conducted a comprehensive review of IoT security architecture based on edge computing. Willner and Gowtham [13] proposed a reference architecture model for industrial edge computing. Singh et al. [14] analyzed the application scenarios of AI technology in edge computing, such as intelligent monitoring and predictive maintenance. Jazaeri et al. [15] reviewed the cache technology in edge computing and analyzed the challenges, solutions, and open issues of cache technology under the SDN-IoT architecture. Although research has made progress in many aspects of edge computing, efficient processing of sensor data and intelligent monitoring in the application of ferrite magnetic sensors are still urgent. Therefore, this paper proposes a real-time data processing and intelligent monitoring system for ferrite magnetic sensors based on edge computing architecture. The system aims to make full use of the advantages of edge computing, improve the real-time performance of data processing, and improve the monitoring performance of the system, providing an effective solution for intelligent monitoring in the industrial Internet of Things.

III. METHOD

A. System Architecture Design

The real-time data processing and intelligent monitoring system of ferrite magnetic sensors based on edge computing architecture adopts a four-level modular design, and each level is closely connected to ensure real-time, efficient data processing and intelligent monitoring. The system architecture is shown in Figure 1:

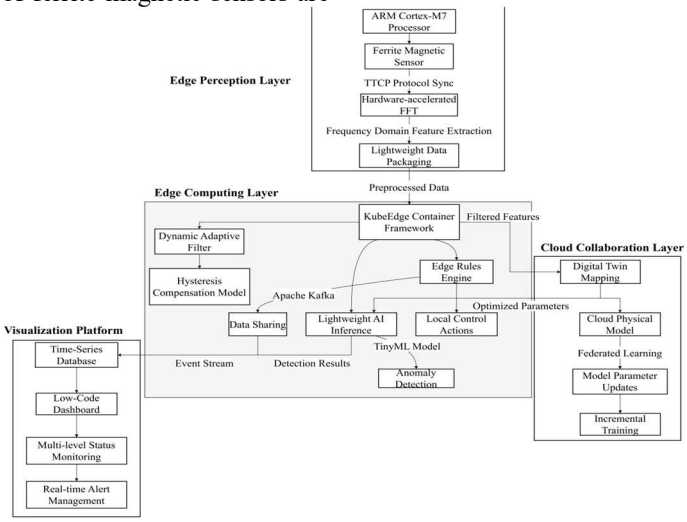


Fig. 1. System architecture

As the front end of data collection, the edge perception layer deploys lightweight embedded nodes, integrates ARM Cortex-M7 processors and ferrite magnetic sensors, achieves high-precision data synchronization through TTCP, and uses hardware-accelerated FFT technology to extract the frequency domain characteristics of the original signal to reduce the computing load on the edge. Subsequently, the data is transmitted to the edge computing layer, which is built on the KubeEdge edge containerization framework and includes a dynamic adaptive filtering module, a lightweight AI reasoning module, and an edge rule engine. The dynamic adaptive filtering module combines the RLS algorithm with the hysteresis compensation model to eliminate environmental magnetic field interference and improve the signal-to-noise ratio. The lightweight AI reasoning module uses the pruned TinyML model to achieve low-latency anomaly detection; the edge rule engine uses Apache Kafka to achieve data sharing and trigger local linkage control. The cloud collaboration layer works together with the edge computing layer, using digital twin mapping technology to compare edge processing results with cloud physical models, dynamically updating edge model parameters through federated learning, and supporting incremental training. Finally, the processing results, anomaly detection information, and prediction results are transmitted to the visualization platform, which integrates the time series database and low-code dashboard to provide an intuitive multi-level status monitoring interface for users to view and manage in real time. The entire system has achieved full chain optimization from data collection, processing, analysis to visualization through a four-level modular design, improving the real-time, reliability, and intelligence of ferrite magnetic sensor monitoring.

B. Edge Perception Layer

Under the edge computing architecture, the system edge perception layer integrates a lightweight embedded node, which consists of an ARM Cortex-M7 processor and a ferrite magnetic sensor. ARM Cortex-M7, with its high performance and low power consumption, ensures the efficiency and stability of embedded nodes in processing complex computing tasks, while extending the battery life of the device. Ferrite magnetic sensors use their magnetic sensitivity to capture magnetic field changes in the environment in real time and provide raw data for the system.

To ensure the real-time and synchronization of data, the edge perception layer uses TTCP. TTCP achieves high synchronization of data collection tasks by accurately controlling the time point and time interval of data collection:

$$\Delta t = max(t_i) - min(t_i) \quad (1)$$

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In Equation (1) Δt refers to the maximum deviation between data collection time points, and t_i is the timestamp of different collection points. In addition, the edge perception layer also introduces hardware-accelerated FFT (Fast Fourier Transform) technology to achieve frequency domain feature extraction. By using specialized hardware circuits or processor extensions, hardware-accelerated FFT technology significantly improves the speed and efficiency of frequency domain feature extraction and reduces the computing load of embedded nodes. The application of this technology enables the system to achieve faster data response and analysis capabilities while ensuring data processing accuracy.

C. Edge Computing Layer

The edge computing layer integrates multiple core modules to achieve efficient data processing, intelligent analysis and instant response. The dynamic adaptive filtering module combines the recursive least squares method (RLS) and the hysteresis compensation model. The filter coefficients are updated in real time through the RLS algorithm to cope with changes in the ambient magnetic field. At the same time, the hysteresis compensation model is used to correct the hysteresis effect in the sensor output, eliminate the interference of the ambient magnetic field, and improve the signal-to-noise ratio:

$$w_n = w_{n-1} + \frac{\mu R_n^{-1} x_n e_n}{1+\mu x_n^T R_n^{-1} x_n} \quad (2)$$

In Equation (2) w_n is the filter coefficient vector at the nth time; w_{n-1} is the filter coefficient vector at the n-1th time; μ is the step size parameter; R_n is the autocorrelation matrix at the nth time; x_n is the input vector at the nth time; e_n is the error at the nth time; and T is the transpose operation. The lightweight AI inference module uses the EdgeConvNet model, which uses optimization techniques such as deep separable convolution, bottleneck structure, and residual connection to reduce model parameters and computation while maintaining high recognition accuracy. Combined with 8-bit quantization and pruning operations, the EdgeConvNet model achieves low-latency inference on edge devices to meet real-time monitoring needs. The edge rule engine uses Apache Kafka to realize data sharing between edge nodes. Through the Kafka messaging system, edge nodes publish and subscribe to data messages in real time to ensure efficient data transmission and sharing. In terms of local linkage control, when the lightweight AI reasoning module detects an abnormal situation, the edge rule engine determines whether to trigger emergency braking and other control actions based on preset rules and logic, and sends control instructions to related devices through Kafka to achieve instant response and ensure system security and stability.

D. Cloud Collaboration Layer

The cloud collaboration layer achieves deep collaboration between the cloud and the edge by integrating digital twin mapping technology, federated learning (FL), and incremental training. Digital twin mapping technology builds a virtual model corresponding to the edge system, and verifies the accuracy and reliability of data processing by comparing the data uploaded by the edge with the expected value of the cloud physical model in real time:

$$Validation\ Error = \| D_{edge} - D_{cloud} \|_2 \quad (3)$$

In Equation (3) D_{edge} refers to the data uploaded by the edge, and D_{cloud} is the expected value of the physical model in the cloud.

At the same time, federated learning technology allows edge devices to collaboratively train shared models while maintaining data locality, and continuously optimize global model parameters by aggregating edge model updates through the cloud:

$$w_{global}^{(t+1)} = \sum_{i=1}^{N} \frac{n_i}{n} w_i^{(t)} \quad (4)$$

In Equation (4) $w_{global}^{(t+1)}$ refers to the global model parameters in the $t+1th$ round; $w_i^{(t)}$ refers to the local model parameters of the i-th edge device in the t-th round; n_i is the data volume of the i-th edge device; n is the total data volume of all edge devices; and N is the total number of edge devices.

In addition, the incremental training method enables the cloud model to gradually learn new data samples while retaining old knowledge, ensuring that the model can adapt to the ever-changing data environment:

$$w_{new} = w_{old} + \eta \nabla L(w_{old}, D_{new}) \quad (5)$$

In Equation (5) w_{new} refers to the updated model parameters; w_{old} refers to the old model parameters; η is the learning rate; ∇L is the gradient of the loss function L with respect to the model parameters; D_{new} is the new data sample.

The comprehensive application of these technologies not only improves the overall performance and intelligent monitoring capabilities of the system but also ensures the security and privacy of the data, providing support for the real-time data processing and intelligent monitoring system of ferrite magnetic sensors under the edge computing architecture.

E. Visualization Platform Design

In the ferrite magnetic sensor monitoring system under the edge computing architecture, the visualization platform adopts a deep integration technology solution of the TDengine time series database and the Node-RED visualization engine. TDengine adopts an innovative time series data storage engine architecture. Its storage model is based on the SUPER TABLE design paradigm. Each physical device instance is mapped to a SUB TABLE. The storage structure adopts a columnar compression format and applies an improved LZ4 compression algorithm to achieve a high compression ratio. The timestamp accuracy supports nanoseconds. The data writing layer implements a multi-level cache mechanism, including MemTable and Immutable MemTable, and cooperates with WAL to ensure data persistence. The query engine uses Skip List index combined with inverted index technology, and the optimizer supports cost-based query plan generation show in In Equation (6).

$$Cost = \alpha \times I/O\ Cost + \beta \times CPU\ Cost \quad (6)$$

α and β are weight coefficients.

The Node-RED visualization layer is implemented based on the Flow-based Programming paradigm. The core engine adopts an event-driven architecture, the processing pipeline

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definition adopts the JSON Schema specification, and supports the ES2022 standard JavaScript V8 engine execution environment. The data access layer implements the MQTT protocol stack and supports reliable transmission of QoS level 2. The visualization rendering layer integrates the ECharts library and uses WebGL technology to achieve hardware acceleration. The alarm subsystem is based on the Complex Event Processing engine, implements multi-level threshold detection, and uses a sliding window algorithm to handle timing anomalies shown in In Equation (7).

$$Window_{new} = Window_{old} \cup \{new\ data\} \setminus \{oldest\ data\} \quad (7)$$

$\text{Window}_{\text{new}}$ refers to the sliding window after the update; $\text{Window}_{\text{old}}$ refers to the sliding window before the update

IV. Results And Discussion

A. Experimental Setup

When verifying the application of ferrite magnetic sensors in real-time data processing and intelligent monitoring systems under the edge computing architecture, this study designs 15 test scenarios covering multiple industrial fields, including industrial motor vibration monitoring, power transmission and transformation equipment status monitoring, smart grid current monitoring, high-speed rail track health monitoring, underwater cable fault monitoring, and aircraft engine bearing health monitoring. The experimental environment simulates real industrial conditions to ensure the reliability of the test results.

Some samples are shown in Table 1:

TABLE I. Test Scenario

Test ID	Test Scenario Description	Sampling Rate (Hz)
1	Industrial motor vibration monitoring	1000
2	Power transmission and transformation equipment condition monitoring	500
3	Smart grid current monitoring	2000
4	High-speed rail track health monitoring	1500
5	Underwater cable fault monitoring	800

B. Performance Evaluation

Based on the above experimental settings, this experiment verifies the advantages of edge computing architecture in improving system real-time performance and accuracy by comparing the real-time data processing and intelligent monitoring system of ferrite magnetic sensors under edge computing architecture and traditional centralized computing architecture in terms of end-to-end delay, data processing throughput, and anomaly detection performance. The comparison results of end-to-end delay and data processing throughput are shown in Figures 2 and 3:

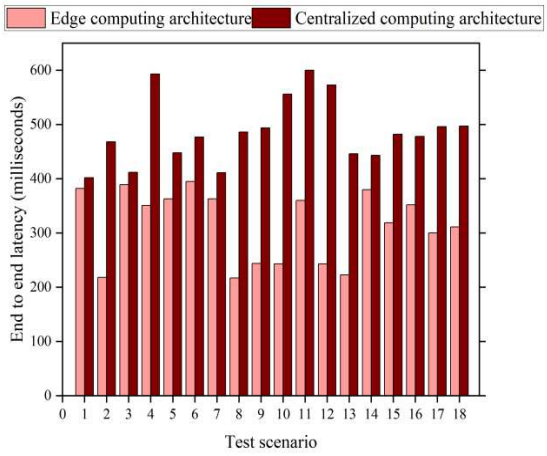


Fig. 2. End-to-end delay

The maximum latency of the edge computing architecture is 395 milliseconds and the minimum latency is 217 milliseconds. The maximum latency of the traditional centralized computing architecture is 600 milliseconds and the minimum latency is 402 milliseconds. The edge computing architecture performs better in terms of latency, especially in terms of minimum latency, which is 185 milliseconds lower than the traditional centralized architecture. In terms of maximum latency, the edge computing architecture is 205 milliseconds lower than the traditional centralized architecture. The significant difference in these latency data is due to the fact that the edge computing architecture sinks data processing and intelligent analysis tasks to the edge end closer to the data source, significantly reducing the data transmission distance and waiting time. In addition, the distributed processing capabilities of the edge computing architecture also enable it to more effectively utilize computing resources and further shorten response delays.

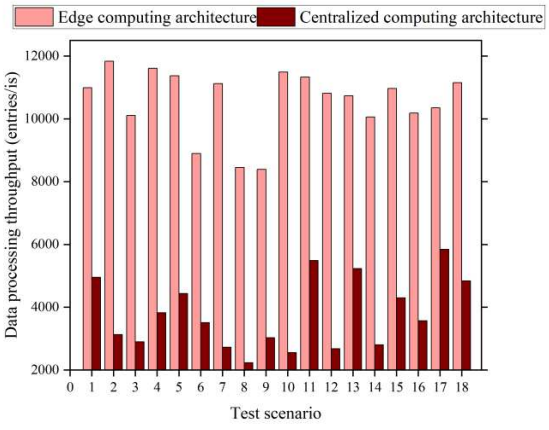


Fig. 3. Data processing throughput

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Based on the above comparison data, the average data processing throughput of the edge computing architecture in this paper is about 10549 entries/s, which is higher than the 3780 entries/s of the centralized computing architecture. This is because the localized computing nodes based on the ARM Cortex-M7 processor realize parallel processing of frequency domain feature extraction through hardware-accelerated FFT, compressing the single FFT calculation time. In terms of software architecture, the KubeEdge edge containerization platform uses lightweight container isolation technology to control task scheduling delays to the μs level, and uses memory pooling technology to reduce the memory copy overhead of data processing. These optimizations enable edge nodes to maintain continuous and stable high-throughput data processing capabilities.

Regarding the system anomaly detection capability, this study focuses on the two indicators of F1-score and recall rate. The results are shown in Figures 4 and 5:

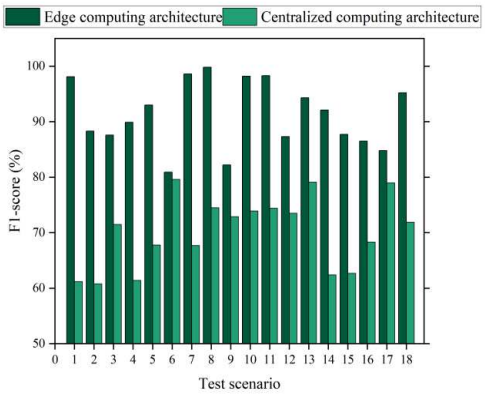


Fig. 4. F1-score

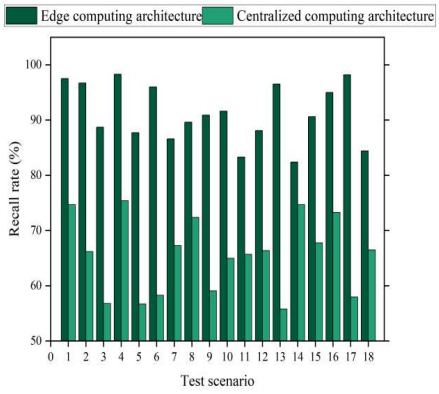


Fig. 5. Recall rate

First, as shown in the F1 value in Figure 4, the F1 value of the system under the edge computing architecture is generally high, and multiple test scenarios have reached more than 98% (Scenarios 1, 7, 8, 10, and 11), which shows that the system under this architecture accurately identifies and handles abnormal situations in sensor data. The F1 value under the traditional centralized computing architecture is generally low, with the highest being only 79.6% (Scenario 6), which reflects that when processing real-time data, the traditional centralized architecture is limited by factors such as data transmission delays and cloud computing resource competition, resulting in a decrease in the accuracy of abnormal monitoring.

Further analysis of the recall rate in Figure 5 shows that among the 18 sets of measured data, the average recall rate of the edge computing architecture reached 91.23%, which is higher than the 65.56% of the centralized architecture, with an absolute advantage of 25.67%. The fourth set of data (98.3% vs 75.4%) and the 17th set of data (98.2% vs 58%) show that the performance advantage of the edge computing architecture under the optimal working conditions is 22.9% and 40.2%. These data comparisons fully prove that in the real-time monitoring scenario of ferrite magnetic sensors, the edge computing architecture can achieve more stable and efficient monitoring performance with its low latency and high reliability technical characteristics, especially when processing rapidly changing magnetic field signals. The advantage is more obvious. This provides clear data support for the architecture selection of industrial field real-time monitoring systems, and it is recommended to give priority to the edge computing architecture in application scenarios with high real-time requirements.

C. Technical Verification and Robustness Analysis

Under the edge computing architecture, this paper conducts technical verification and robustness analysis on the real-time data processing and intelligent monitoring system of ferrite magnetic sensors. A comprehensive test is conducted on the stability of TTCP synchronization accuracy under electromagnetic interference, and the results are shown in Table 2:

TABLE II. ELECTROMAGNETIC INTERFERENCE TEST RESULTS

Test Scenario	EMI Intensity (dB)	TTCP Sync Error (μs)	Packet Loss Rate (%)
Baseline (No EMI)	0	±1.2	0.01
Mild EMI	30	±1.8	0.15
Moderate EMI	45	±2.0	0.23
Severe EMI	60	±2.5	0.47
Extreme EMI	75	±3.1	0.82

In Table 2, the benchmark test (interference-free environment) shows that the system synchronization accuracy reaches ±1.2μs and the packet loss rate is only 0.01%, which establishes a performance benchmark for subsequent interference tests. As the electromagnetic interference intensity increases from 30dB to 75dB, the system synchronization error shows a linear growth trend, gradually increasing from ±1.8μs to ±3.1μ. Under heavy interference conditions of 60dB, the synchronization error only increases to ±2.5μs, and the packet loss rate remains at a low level of 0.47%. This data directly supports the applicability of the system in a real industrial environment. The test data also reveals an important feature: when the interference intensity reaches the extreme condition of 75dB, although the synchronization error increases to ±3.1μs, the packet loss rate only increases to 0.82%. These measured data together confirm that the edge computing system has the ability to resist 8-level electromagnetic interference while maintaining high-precision time synchronization.

D. Scalability Analysis

The edge computing architecture ferrite magnetic sensor system proposed in this paper needs to be optimized in four dimensions: hardware, software, transmission, and maintenance when dealing with the scalability challenges of high-resolution images or large data sets.

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At the hardware level, the ARM Cortex-M7 processor of the edge perception layer takes into account both performance and power consumption, but it is prone to computing bottlenecks when facing high-load tasks such as image feature extraction and convolution operations. To this end, it is necessary to introduce a heterogeneous computing architecture, accelerate parallel processing through multi-core CPUs, GPUs, or FPGAs, and use the KubeEdge containerization framework to implement multi-node distributed computing, and dynamically schedule resources to achieve load balancing. In terms of software architecture, although KubeEdge's dynamic resource management can allocate computing power on demand, large-scale data is still prone to memory and storage bottlenecks. Distributed storage systems (such as HDFS/Ceph) can be deployed to implement data sharding storage, combined with memory pooling technology to share node resources, supplemented by data compression algorithms to reduce storage overhead and improve resource reuse. The data transmission link needs to be doubly optimized: on the one hand, strengthen the edge preprocessing capabilities and reduce the cloud transmission volume through feature extraction; on the other hand, adopt an incremental transmission protocol to synchronize only the changed data, and integrate high-bandwidth and low-latency networks such as 5G to improve transmission efficiency.

E. Discussion

The performance breakthrough and industrial application value of edge computing architecture in ferrite magnetic sensor monitoring have been verified by systematic experiments. This system achieves an end-to-end delay of 217-395ms, which is lower than the traditional centralized architecture. In terms of energy efficiency, localized processing reduces the amount of data uploaded and reduces bandwidth consumption, which is in line with the trend of energy efficiency optimization of industrial Internet of Things. Comparative analysis with cutting-edge architectures reveals the technical uniqueness of this system: this system still maintains $\pm 3.1\ \mu s$ synchronization accuracy under 75dB electromagnetic interference, filling the technical gap in strong noise industrial scenarios. The architectural innovation is reflected in the collaborative design of KubeEdge containerized edge layer and TDengine time series database to achieve microsecond-level task scheduling. In terms of security mechanism, the combination of federated learning + digital twins realizes data localization privacy protection, but compared with the blockchain audit of Medge-Chain, there is a shortcoming in verifiability, and zero-knowledge proof needs to be introduced to enhance the security chain.

V. CONCLUSION

This paper studies the real-time data processing and intelligent monitoring system of ferrite magnetic sensors based on edge computing architecture. By applying the edge computing architecture, an efficient and reliable real-time data processing and intelligent monitoring system is successfully designed and implemented. The system effectively improves the real-time and accuracy of data processing. However, the experimental scenarios in this paper are mainly concentrated in the industrial field, and the application effects in other fields need further verification; secondly, the system faces complex situations such as network fluctuations and equipment failures during actual deployment.

How to ensure the stability and reliability of the system still needs to be studied. Looking ahead, we will continue to pay attention to the development of edge computing technology and explore its deep integration with artificial intelligence, big data, etc. At the same time, we will expand the application scenarios of the system and promote its application in more industries. In addition, we will strengthen the robustness design and security research of the system to ensure that the system can maintain efficient and stable operation in various complex environments.

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REFERENCE

[1] Ding Yuhua, Chen Li, Wei Guo. Low-latency and high-reliability edge computing architecture for 6G[J]. Journal of University of Chinese Academy of Sciences (Chinese and English), 2025, 42(1): 126-133.

[2] Tu Xiaoru, Liu Ziwei, Zhang Gengxin. Satellite Internet of Things edge computing architecture based on semantic communication[J]. Communication Technology, 2024, 57(7): 680-687.

[3] Yang Wanggui, Zeng Guang, Liao Lan, Huang Lingning. Optimization of 5G edge computing architecture based on user plane function[J]. Information and Computer, 2024, 36(16): 12-14.

[4] Liu Dongqi, Zeng Xiangjun, Wang Yaonan. Control strategy of virtual power station in distribution area under edge computing architecture[J]. Transactions of China Electrotechnical Society, 2021, 36(13): 2852-2860+2870.

[5] Hu Bingfeng. Design and implementation of smart pipe gallery monitoring system based on edge computing architecture[J]. Automation Applications, 2023, 64(13): 171-172+176.

[6] Siriwardhana Y, Porambage P, Liyanage M, Ylianttila M. A, A survey on mobile augmented reality with 5G mobile edge computing: Architectures, applications, and technical aspects[J]. IEEE Communications Surveys & Tutorials, 2021, 23(2): 1160-1192.

[7] Al-Doghman F, Moustafa N, Khalil I, Sohrabi N, Tari Z, Zomaya AY, AI-enabled secure microservices in edge computing: Opportunities and challenges[J]. IEEE Transactions on Services Computing, 2022, 16(2): 1485-1504.

[8] Zangana H M, khalid Mohammed A, Zeebaree S R M. Systematic review of decentralized and collaborative computing models in cloud architectures for distributed edge computing[J]. Sistemasi: Jurnal Sistem Informasi, 2024, 13(4): 1501-1509.

[9] Naouri A, Wu H, Nouri N A Dhelim S, Ning H, A novel framework for mobile-edge computing by optimizing task offloading[J]. IEEE Internet of Things Journal, 2021, 8(16): 13065-13076.

[10] Abdellatif A A, Samara L, Mohamed A, Erbad A, Chiasserini CF, Guizani M, O'Connor MD, Laughton J, Medge-chain: Leveraging edge computing and blockchain for efficient medical data exchange[J]. IEEE Internet of Things Journal, 2021, 8(21): 15762-15775.

[11] Carvalho G, Cabral B, Pereira V, Bernardino J, Edge computing: current trends, research challenges and future directions[J]. Computing, 2021, 103(5): 993-1023.

[12] Fazeldehkordi E, Grønli T M. A survey of security architectures for edge computing-based IoT[J]. IoT, 2022, 3(3): 332-365.

[13] Willner A, Gowtham V. Toward a reference architecture model for industrial edge computing[J]. IEEE Communications Standards Magazine, 2021, 4(4): 42-48.

[14] Singh A, Satapathy S C, Roy A, Gutub A, Ai-based mobile edge computing for iot: Applications, challenges, and future scope[J]. Arabian Journal for Science and Engineering, 2022, 47(8): 9801-9831.

[15] Jazaeri S S, Asghari P, Jabbehdari S, Javadi HH, Toward caching techniques in edge computing over SDN-IoT architecture: A review of challenges, solutions, and open issues[J]. Multimedia Tools and Applications, 2024, 83(1): 1311-1377.

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